

Skills Summary

Biased Samples and Mismatched Denominators

Use this reference while completing your worksheet or when studying.

Getting the Denominator Right

Rule: the bottom of a probability counts *every* equally likely outcome the draw could produce — not one of the groups inside it.

$$P(\text{event}) = \frac{\text{favourable outcomes}}{\text{all outcomes}}$$

Tell: “5 red and 7 blue” gives a denominator of 12, never 7. If your fraction is bigger than 1, the denominator is certainly wrong.

Example: A tin holds 6 red and 9 blue pegs. One is drawn at random. Find the probability of red.

Step 1. The draw chooses from every peg in the tin, so the denominator is the total $6+9 = 15$, not the 9 blue ones.

Step 2. Six of the fifteen pegs are red, so the probability is $\frac{6}{15}$, which reduces to **2/5**.

Try it: A drawer holds 7 black socks and 13 white socks. Find the probability of drawing a black sock.

check: 7/20

Spotting a Sample that Was Never Random

Rule: a sample is random only when every member of the population had the same chance of being chosen. If not, correct arithmetic still produces a useless estimate.

Tells: the group was gathered *because* of the thing being measured (cinema-goers asked about cinemas), or people put themselves in by replying, volunteering or turning up.

Example: At a football match, 24 of the 30 people asked said football is their favourite sport. A student scales this to the town’s 6000 residents. What does the arithmetic give, and why is it wrong?

Step 1. Scaling a sample means multiplying the population by the sample fraction, so do that first and then ask where the sample came from.

Step 2. $\frac{24}{30} \times 6000 = 4800$ residents — correct arithmetic, worthless estimate, because everyone asked was already at a football match.

Try it: Outside a library, 18 of 20 people asked said they read daily. What would this sample estimate for a town of 5000, and why should nobody believe it?

check: 4500, and the sample is readers

When Something Is Removed, Recount Both Numbers

Rule: once an item is taken away and not replaced, the probability is computed from what is *left*. If the item removed was one of the favourable ones, the top drops by one **and** the bottom drops by one. If it was not favourable, only the bottom drops.

Example: A jar holds 5 red and 7 green beads. One red bead is taken out and kept. Find the probability that the next bead drawn is red.

Step 1. The bead removed was red, so it comes off both counts — recount before writing the fraction.

Step 2. Four red beads are left out of 11 beads in total, so the probability is $\frac{4}{11}$.

Try it: A box holds 8 blue and 12 white tiles. One blue tile is removed and kept. Find the probability that the next tile drawn is blue.

check: $\frac{6}{19}$

(!) **Watch out:** a probability above 1 is impossible, and it almost always means the denominator counted the wrong group. Check the bottom of the fraction before you check anything else.